

The median nerve carries signals from touch sensors in your hand to your brain.

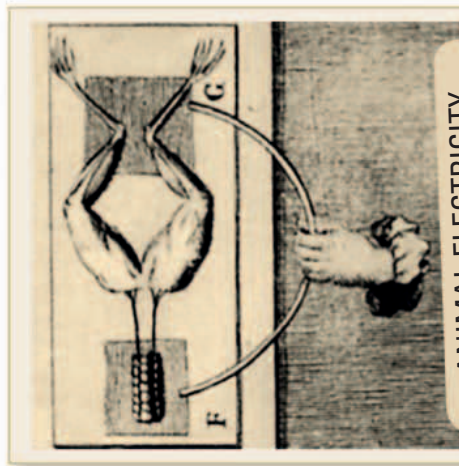
Your ulnar nerve carries signals to muscles in your hand.

The femoral nerve activates muscles that straighten the leg.

The sciatic nerve is the longest in the body, running from the spine to the foot.

The tibial nerve flexes the calf and foot and provides the push that allows you to walk.

The peroneal nerve moves the toes up and down.



ANIMAL ELECTRICITY

The first person to realize that the human body uses electricity was an Italian doctor named Luigi Galvani. In the 1780s, Galvani discovered that he could make a frog's leg twitch by jolting the muscle with electricity. He concluded that animal muscles are activated by electrical signals, which he called "animal electricity."



HIGH-SPEED COMMUNICATION

Your nerves transmit signals to and from the brain with great speed. This ping-pong player is able to hit a ball hurtling toward him because signals are traveling along his nerves at up to 220 mph (350 kph). In a split second, signals zoom from his eyes to his brain to warn that the ball is on its way. His brain then sends signals to his arm muscles, controlling his return shot with amazing precision.

FAST FACTS

- An average adult human brain weighs 2½–3 lb (1.3–1.4 kg).
- A sperm whale's brain—the world's biggest animal brain—weighs 18 lb (8 kg).
- The longest human nerves are more than 3 ft (1 m) long and stretch from the lower part of the spinal cord to the toes.
- A giraffe has nerves more than 10 ft (3 m) long running down its neck.

